

Khai Loong Aw

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Education

- 2024 - **Stanford University**, *Ph.D. Computer Science*
Research on AI, cognitive science, and neuroscience.
- 2020 - 2024 **Singapore Management University**, *B.Sc. Computer Science*
GPA: 4.0/4.0.

Peer-Reviewed Publications

- 2024 Instruction-tuning Aligns LLMs to the Human Brain
K.L. Aw, S. Montariol*, B. AlKhamissi*, M. Schrimpf**, A. Bosselut**
COLM 2024
- 2023 Training language models to summarize narratives improves brain alignment
K.L. Aw, M. Toneva
ICLR 2023 - **Top 25% Notable Paper (Spotlight)**
- 2022 Detecting False Alarms from Automatic Static Analysis Tools
H.J. Kang, **K.L. Aw**, D. Lo
ICSE 2022 - **Distinguished Paper Nominee** (top 19 papers, top 3% of submissions)

Research Experience

- 2024 **Stanford**, *CS PhD student*
• Research Rotation Advisor: Daniel Yamins
- 2023 **EPFL**, *Research Intern (NeuroAI)*
• Advisors: Martin Schrimpf, Antoine Bosselut
• Investigated the similarities between: (1) LLM representations and human brain fMRI activity, and (2) LLM and human behavior. Showed that LLM-brain alignment strongly correlates with model size ($r = 0.95$) and world knowledge ($r = 0.81$). [COLM 2024]
• Contributed LLaMA and T5 models to the open-science Brain-Score Language platform
- 2022 **Max Planck Institute for Software Systems**, *Research Intern (NeuroAI)*
• Advisor: Mariya Toneva
• Designed a novel interpretability method to evaluate brain alignment across various narrative elements (e.g., characters, emotions). Showed that training language models to summarize narratives improves alignment to human brain fMRI activity. [ICLR top 25%]
- 2022 **Singapore Management University**, *Research Assistant (Computer Vision)*
• Advisor: Sun Qianru
• Used Self-Supervised Learning as a pre-training step to improve the performance of existing methods for the computer vision task of Semantic Segmentation
- 2021 **Software Analytics Research Group**, *Research Intern (Machine Learning)*
• Advisor: David Lo
• Used AI interpretability methods to identify data leakage issues with state-of-the-art ML models for detecting software bugs. [ICSE Distinguished Paper Nominee]
- 2017 **National University of Singapore**, *Research Assistant (Engineering)*
• Advisor: Dan Zhao
• Designed experiments to synthesize novel compounds for carbon capture applications

Industry Experience

- 2023 **GIC Singapore, NLP Project**
• Built pipeline of text-to-SQL and retrieval-augmented generation LLMs, allowing employees to use English questions to query global stock market databases
- 2022 **GovTech Singapore, AI Engineer Intern (Computer Vision)**
• Improved the processing speed of state-of-the-art algorithms for Multi-Object Tracking, i.e., tracking the motion of multiple objects or humans within a video sequence

Presentations

- 2023 **Oral, ICLR 2023**
- 2023 **Talk, ML Tea @ Max Planck Institute for Software Systems**

Awards

- 2024 **Stanford School of Engineering Fellowship**
- 2024 **IMDA Gold Medal Award** - top student in B.Sc. CS graduating cohort
- 2022 **Best Poster Presentation**, MPI-SWS Institute Retreat 2022
- 2022 **Outstanding Second Year Student Award** - top student in Year 2 of CS undergrad
- 2021 **Outstanding First Year Student Award** - top student in Year 1 of CS undergrad
- 2020 **Global Impact Scholarship** - undergrad university's top full-ride scholarship

Hackathons

- 2021 **1st place**, SMU Legal Innovation and Technology (LIT) Hackathon
- 2021 **3rd place**, SMU Ellipsis Tech Series Hackathon presented by Goldman Sachs
- 2021 **Dentons Rodyk Access to Justice Award**, SMU LIT Hackathon
- 2021 **People's Choice Award**, Asia-Pacific Legal Innovation and Technology Association
- 2021 **Tech-cess to Justice Award**, Asia-Pacific Legal Innovation and Technology Association
- 2021 **Legal Tech Entrant Award**, Asia-Pacific Legal Innovation and Technology Association
- 2020 **Best College Hack**, Hack-a-Solution international hackathon

Service

- 2023 - now **Reviewer**
NeurIPS 2024 (Top Reviewer)
Queer in AI Workshop @ NeurIPS 2024, NAACL 2024
Workshop on Representational Alignment @ ICLR 2024
Workshop on Unifying Representations in Neural Models @ NeurIPS 2023
Workshop on Instruction Tuning and Instruction Following @ NeurIPS 2023
- 2021 - now **Mentor**
Stanford CS PhD Student-Applciant Support Program (SASP) 2024
Stanford CS Undergraduate Mentoring 2024
Stanford Graduate Application Mentorship Program (GradAMP) 2024
QueerInAI Graduate Application Mentorship Program 2024
SMU "Pay It Forward" Mentorship Program 2024
SMU Data Associate Program 2023 - mentored 4 students on their ML project
SMU Data Associate Program 2021
- 2021 - 2022 **Vice-President, SMU BIA**
Led team of 23 directors to organize Machine Learning workshops & events for a club with 1000+ students; allow non-computing students to gain hands-on ML experience

- 2020 - 2022 **Organizer, SMU BIA**
Workshop on ML Deployment
Workshop on Deep Learning in NLP
Workshop on Interview Preparation for ML and Data Engineers
Workshop on Interview Preparation for Business and Data Analysts
Other events: Data Science CTF Hackathon, Industry Mentor Sharing Session, Project Polish Program, and projects between SMU BIA and industry partners
- 2017 - 2021 **Volunteer**
Overseas Community Service Project *Vikasa* - taught students in Jaipur, India
Meet-the-People Sessions - write letters to help people get financial aid
Willing Hearts - help in soup kitchen